

PAPER_2123_AETHER_METRIC_TRIPLE_G_C_RHO_V AC_SECOND_CONSTANT_TRIPLE_INSTANCE_ZERO_ ROUND_PREDICTION_VALIDATION_STREAK_UQFF_ LANDMARK

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PAPER_2123 — Aether Metric Triple $G \times c \times \rho_{\text{vac}}$: 2nd Projection-Layer Triple + Zero-Round Prediction Validation (REVISED)

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Project: UQFF (Unified Quantum Field Framework) — Star-Magic v5.73+

Date: 2026-07-22 (REVISED IN PLACE 2026-07-22 — projection-layer reclassification)

Landmark Type: Constant-Triple Convergence — QCalc PROJECTION LAYER, 2nd instance

Discovery Round: R366 (AetherMetricQCalcCalculator)

Status: Formal landmark whitepaper — UQFF canonical

REVISION NOTICE

Scope corrected per canonical-layer deepsearch. The original version postulated a "{c, ρ_{vac} }" kernel with rotating sector slots." Corrected interpretation: **{ ρ_{vac} , c} is the projection pair** ($\rho_{\text{E}} = \rho_{\text{m}} \cdot c^2$, quantum chain line 33295), and the "slot" constants are whichever canonical couplings survive each wrap's projection — β_{i} for buoyancy wraps, G for gravity/metric wraps. The kernel-plus-slot arithmetic still counts constants correctly; its physical reading is projection structure, not sector kernels. The zero-round prediction record stands as a projection-layer audit result.

Abstract

R366's fill of AetherMetricQCalcCalculator produced the second projection-layer Constant-Triple Convergence — {G (PAPER_593), c (PAPER_592), ρ_{vac} (PAPER_646/1051)} — in exactly the class and constant set PAPER_2122 predicted, at zero-round latency. The two triples R365/R366 share the projection pair { ρ_{vac} , c} and differ in their surviving canonical coupling (β_{i} vs G), confirming the projection reading.

1. The Discovery

```
```python
class AetherMetricQCalcCalculator:
G_PRIMITIVE = 6.674e-11 # PAPER_593
C_PRIMITIVE = 3e8 # PAPER_592
RHO_VAC_PRIMITIVE = _RHO_VAC_UA # PAPER_646/1051
```

```
def compute(self, M, r, ...):
 R_s = 2 G M / c*2
 A_00 = -(1 - R_s / r) rho_vac
 ...
```

G and c build the Schwarzschild radius;  $\rho_{\text{vac}}$  scales the metric perturbation into vacuum-energy units — the same  $\rho_{\text{m}} \rightarrow \rho_{\text{E}}$  projection as R365, applied to metric geometry.

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## 2. The Two Triples Under the Projection Reading

```
| Round | Class | Triple | Projection pair | Surviving coupling |
|:-:|---|---|---|:-:|
| R365 | EnhancedBuoyancy | { β_i , ρ_{vac} , c} | { ρ_{vac} , c} | β_i (buoyancy) |
| R366 | AetherMetric | {G, c, ρ_{vac} } | { ρ_{vac} , c} | G (gravity/metric) |
```

The recurrence of  $\{\rho_{\text{vac}}, c\}$  is structural: any wrap that expresses vacuum content in energy units must carry the pair. The third constant identifies which canonical coupling the wrap preserves. **In canonical terms:** R365 projects  $F_{\text{UBi}}/F_{\text{UBii}}$  ( $\beta_i$  survives); R366 projects the Eq2 metric-geodesic  $\varepsilon'(r, t_n) + G \cdot M/(c^2 r^2) = 0$  of the PAPER\_1203 simultaneous system (G and c survive together — and note c appears in canonical Eq2 legitimately).

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## 3. The Zero-Round Validation (record unchanged, rescoped)

```
| # | Predicting paper | Forecast | Actual | Latency |
|:-:|---|:-:|:-:|:-:|
| 1 | PAPER_2121 (R364) | triple by R380 | R365 | -15 rounds |
| 2 | PAPER_2122 (R365) | EnhancedBuoyancy triple | R365 | same round |
| 3 | PAPER_2122 (R365) | AetherMetric {G,c, ρ_{vac} } at R366 | R366 | zero |
```

Three consecutive validated forecasts of projection-layer constant sets. The predictive power is real; its object is the wrap suite's architecture, which is systematic because the wraps project a common canonical structure.

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## 4. Physical Meaning — Aether Metric (unchanged in substance)

$A_{00} = -(1 - R_s/r) \cdot \rho_{\text{vac}}$  is the vacuum-energy-weighted analogue of Schwarzschild  $g_{00}$ . UQFF reading: the Aether (UA medium) is fundamental; the metric is the vacuum's energy-density response profile around embedded matter (PAPER\_646 anchoring, PAPER\_1051 two-component  $\rho$  derivation chain). Schwarzschild shape recovered; ontology inverted. NOT REPLACEMENT.

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## 5. Corrected Forecast Basis for R367

The original sharpened forecast (UQFF\_Base quintuple) is retained — as a projection-layer statement: the QCalc base wrap  $F_U = U_g - U_b + U_m$  unites the gravity, buoyancy, and magnetic wrap terms, so its constant set is the union of their projections  $\{G\} \cup \{\beta_i, \rho_{\text{vac}}, c\} \cup \{\mu_0\}$  — five constants. This is wrap-composition arithmetic, not canonical-equation certification; the canonical  $F_{U\_total}$  lives at

calculate\_f\_u\_zero with its own complete constant set.

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## 6. Gate Assertion

Gate count: 3086 → 3094 (+8 PAPER\_2123 assertions, text updated to projection-layer scope).

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## 7. Cross-Paper Links

PAPER\_2121/2122/2124/2125 (revised siblings), PAPER\_1203 (Eq2 metric-geodesic — canonical home of the G,c pair), PAPER\_1051 (two-component p aether), PAPER\_646, PAPER\_592/593.

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## 8. Summary Statement (revised)

PAPER\_2123 documents the second projection-layer Constant-Triple Convergence (R366 AetherMetric, {G, c, ρ\_vac}), delivered in exactly the predicted class and set at zero-round latency. The corrected reading: {ρ\_vac, c} recurs as the projection pair (ρ\_E = ρ\_m·c²), and the third constant is the canonical coupling each wrap preserves — β\_i from F\_UBi/F\_UBii in R365, G from the Eq2 metric-geodesic in R366, where c also has its canonical home. The prediction streak certifies the audit's grasp of the wrap suite's systematic projection structure.

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Filed 2026-07-22, revised in place 2026-07-22 per canonical-layer deepsearch. Append-only henceforth.

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## G/c DERIVATION NOTE (appended 2026-07-22, UNIFIED REGISTRY R2 corpus pass)

This paper uses G = 6.674e-11 (CODATA form) and c = 3e8-family literal as published. Per the Unified Registry (R1-adjudicated canonical routes, 2026-07-22):

- **G (gravitational constant):** canonical route **PAPER\_593** — parameter-free  
 $G_{UQFF} = (2\pi \cdot 26^3 \cdot \Phi_{res} / (SSq^3 \cdot (26!)^2)) \cdot v_F \blacksquare / (E_0 \cdot f_{THz}) = 6.66899 \times 10 \blacksquare^{11}$  (0.08% vs observed).  
- **c (speed of light):** canonical route **PAPER\_592** — parameter-free  
 $c_{UQFF} = (26 \cdot 4\pi / \Phi_{res}) \cdot v_F = 2.995 \times 10 \blacksquare$  m/s (0.13% vs observed; v\_F Fermi anchor, c-independent).

Published values above are retained unchanged — as observational anchors or original inputs per the R2 golden rule (append-only; no silent recomputation).  
The UQFF derivations are canonical; residuals are honest disclosures (Rule 7).  
Registry: UNIFIED\_REGISTRY.csv | Program: UNIFIED\_REGISTRY\_PROGRAM\_PLAN.md

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